



Producción de ondas gravitatorias durante el Recalentamiento: análisis teórico y simulaciones con *CosmoLattice*

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Resumen

Aca va el resumen

Palabras clave:

Agradecimientos

Acá van los agradecimientos

.

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Capítulo 1

Introducción

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1.1. Inflación

1.2. Recalentamiento

Capítulo 2

Ondas Gravitacionales

2.1. Detección

2.2. Mecanismos de Producción

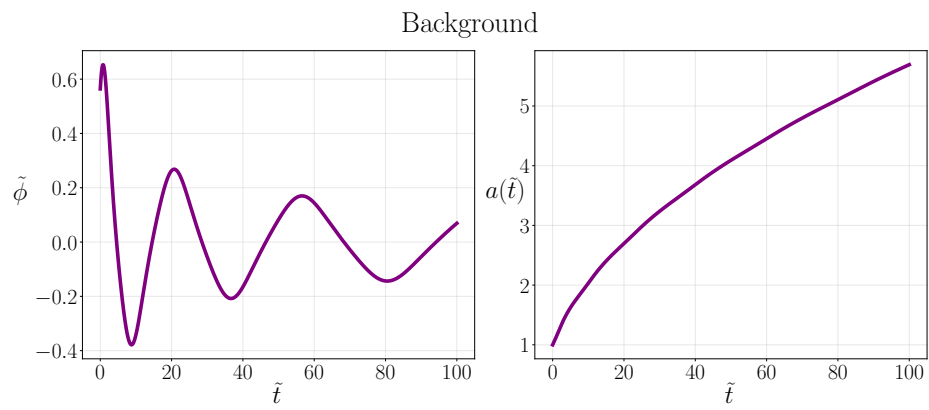
Capítulo 3

Evolución lineal de los campos

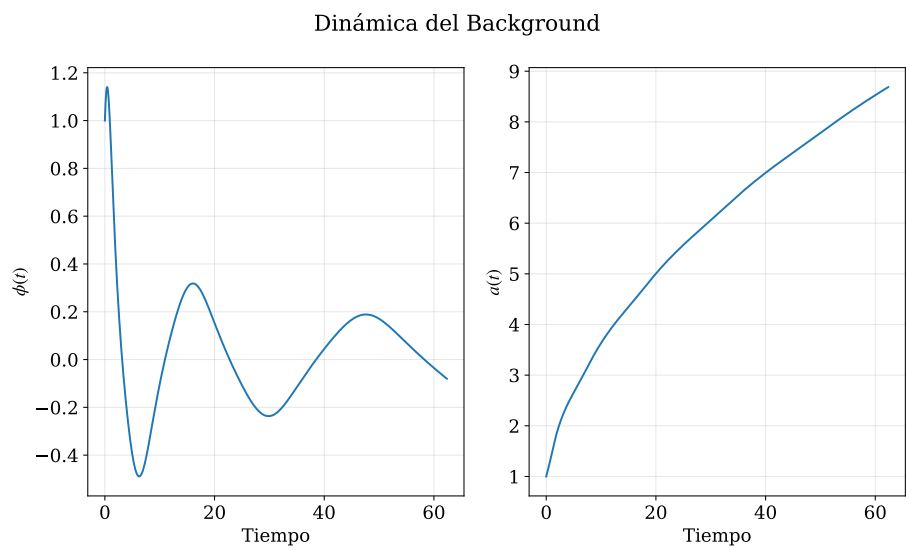
3.1. Resonancia Paramétrica

3.2. Análisis de Floquet

3.3. Producción lineal de ondas gravitacionales

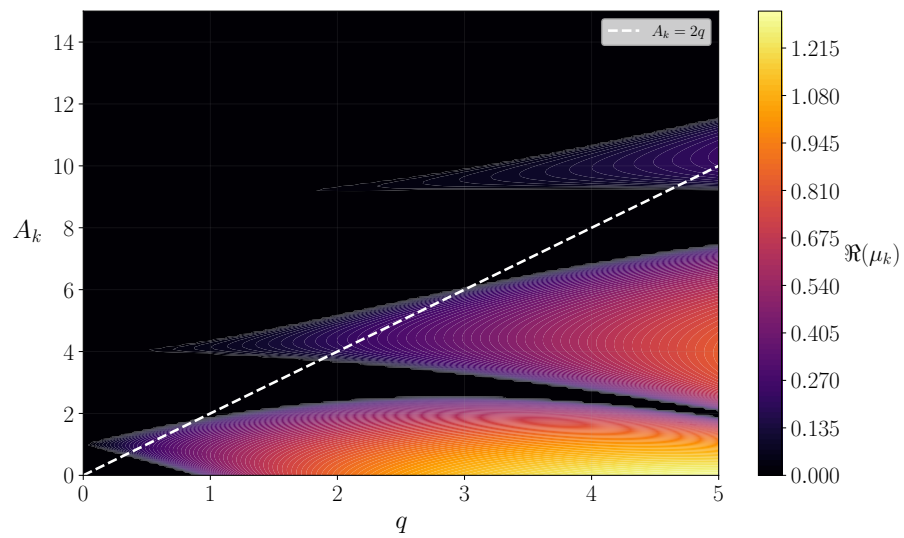


(a) $m^2\phi^2$

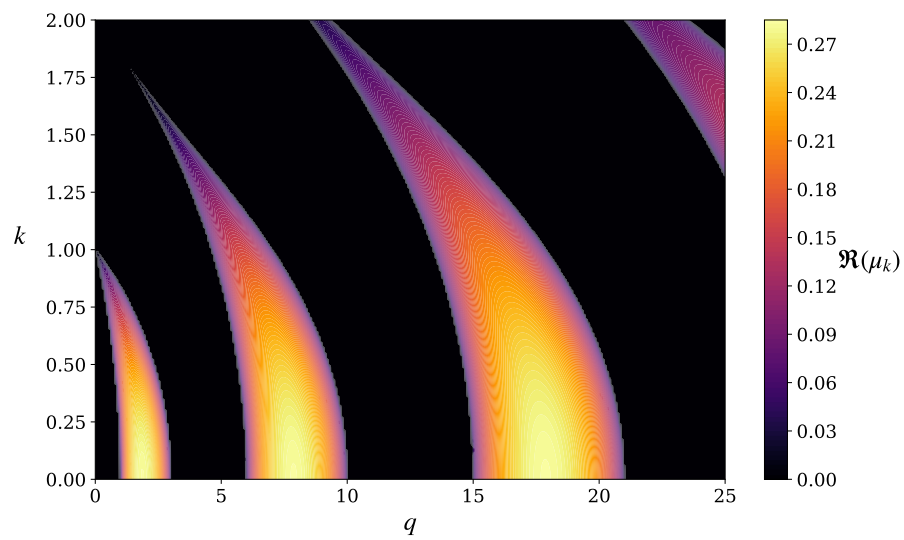


(b) $\lambda\phi^4$

Figura 3.1

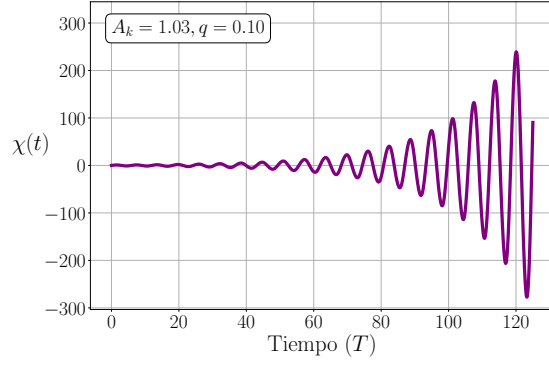


(a) $m^2\phi^2$

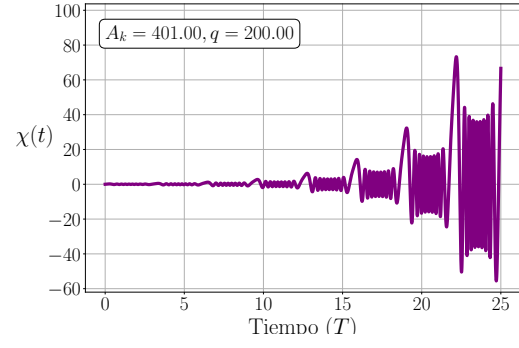


(b) $\lambda\phi^4$

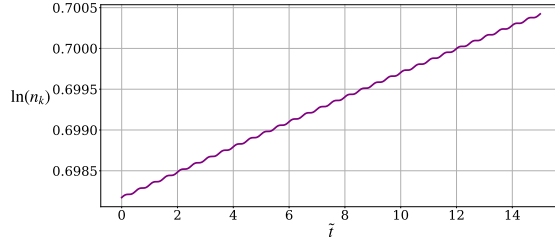
Figura 3.2



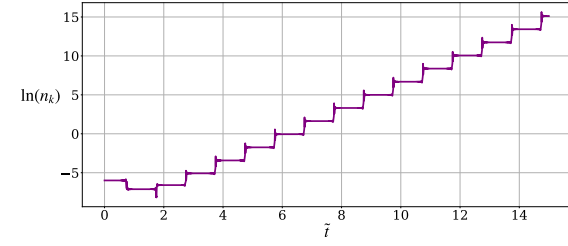
(a) Narrow resonance: amplitud



(b) Broad resonance: amplitud

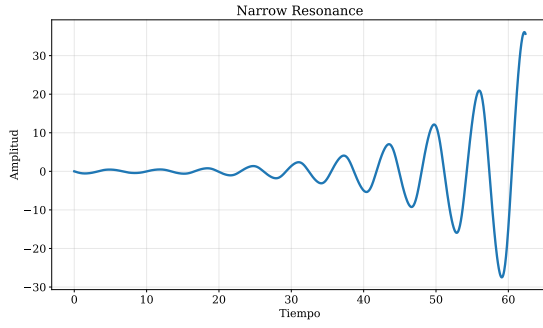


(c) Narrow resonance: número de ocupación

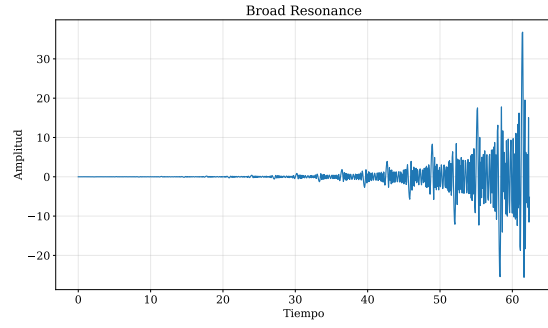


(d) Broad resonance: número de ocupación

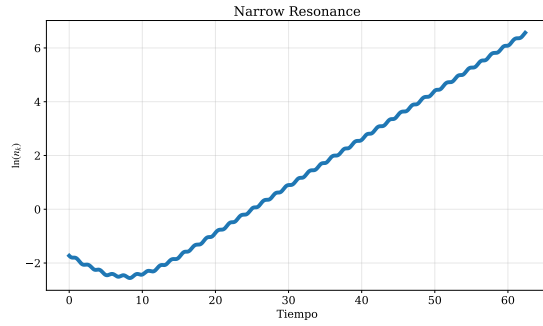
Figura 3.3



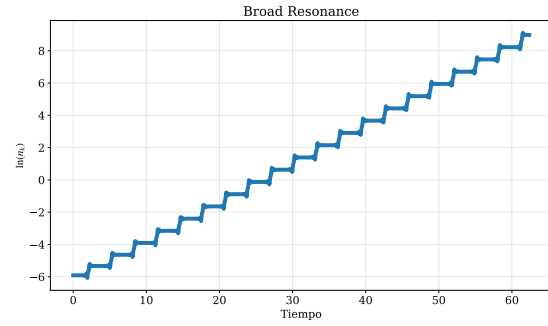
(a) Narrow resonance: amplitud



(b) Broad resonance: amplitud



(c) Narrow resonance: número de ocupación



(d) Broad resonance: número de ocupación

Figura 3.4

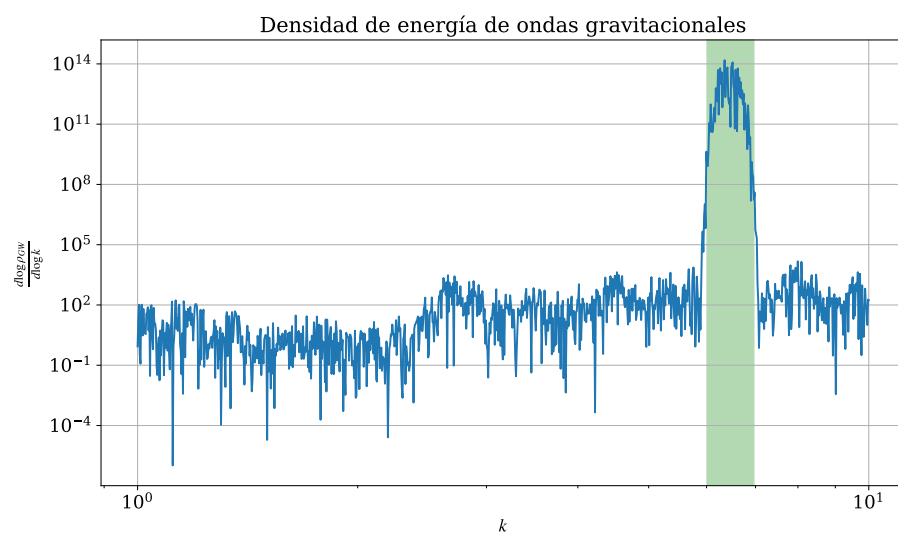


Figura 3.5

Capítulo 4

CosmoLattice

4.1. ¿Qué es?

4.2. Estudio del potencial $\lambda\phi^4$ a orden lineal

4.3. Evolución del background

4.4. Comparación de modelos con $\omega_{eff} = 1/3$

4.5. Comparación de modelos con $\omega_{eff} = 0$

Capítulo 5

Conclusiones

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Apéndice A

Ánàlisis en la estabilidad de modelos en *CosmoLattice*

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